

Name: _____

Solutions

Date: _____



SAFETY BAY
SENIOR HIGH SCHOOL
imagine believe achieve

Year 10 Mathematics

Investigation 3, 2015

Topic – Division of Polynomials

Validation Section

Total Time: **40 minutes**

Weighting: **10% of the year.**

Equipment: Division of Polynomials Booklet – complete in-class section
No Calculators are to be used

77
= _____ %

Use division methods as detailed in the Booklet to answering the following. Note, marks will be given for **each correct line of work plus the final answer**. Use the **same format** as the examples in the Booklet.

1. $93 \div 5$ (5 marks)

$$\begin{array}{r} 18 \text{ r } 3 \\ 5 \overline{) 93} \\ \underline{5} \quad \checkmark \\ 43 \quad \checkmark \\ \underline{40} \quad \checkmark \\ 3 \quad \checkmark \end{array}$$

18 r 3 ✓

2. $722 \div 3$ (5 marks)

$$\begin{array}{r} 240 \text{ r } 2 \\ 3 \overline{) 722} \\ \underline{6} \quad \checkmark \\ 12 \quad \checkmark \\ \underline{12} \quad \checkmark \\ 02 \quad \checkmark \end{array}$$

240 r 2 ✓

3. $3975 \div 7$ (7 marks)

$$\begin{array}{r} 567 \text{ r } 6 \\ 7 \overline{) 3975} \\ \underline{35} \quad \checkmark \\ 47 \quad \checkmark \\ \underline{42} \quad \checkmark \\ 55 \quad \checkmark \\ \underline{49} \quad \checkmark \\ 6 \quad \checkmark \end{array}$$

567 r 6 ✓

7. Divide $11x^3 + 6x + 2$ by $x - 1$ (10 marks, 6 working, 2 layout, 2 answer)

$$\begin{array}{r}
 11x^2 + 11x + 17 \\
 x-1 \overline{) 11x^3 + 0x^2 + 6x + 2} \\
 \underline{11x^3 - 11x^2} \quad \checkmark \\
 11x^2 + 6x \quad \checkmark \\
 \underline{11x^2 - 11x} \quad \checkmark \\
 17x + 2 \quad \checkmark \\
 \underline{17x - 17} \quad \checkmark \\
 19 \quad \checkmark
 \end{array}$$

setting out $\checkmark$
using $0x^2$ $\checkmark$

$$(11x^2 + 11x + 17)(x - 1) + 19 \quad \checkmark \checkmark$$

8. Show that $x - 2$ and $x - 1$ are both factors of $x^3 + x^2 - 10x + 8$ (8 marks)

$$p(x) = x^3 + x^2 - 10x + 8$$

$$(x - 2) \therefore a = 2 \quad \checkmark$$

$$p(2) = 2^3 + 2^2 - 10 \times 2 + 8 \quad \checkmark$$

$$= 8 + 4 - 20 + 8 \quad \checkmark$$

$$= 0$$

Therefore $(x - 2)$ is a factor $\checkmark$

$$(x - 1) \therefore a = 1 \quad \checkmark$$

$$p(1) = 1^3 + 1^2 - 10 \times 1 + 8 \quad \checkmark$$

$$= 1 + 1 - 10 + 8 \quad \checkmark$$

$$= 0$$

Therefore $(x - 1)$ is a factor $\checkmark$

9. Show that $x+1$ and $x-3$ are both factors of $2x^3 - 5x^2 - 4x + 3$ (8 marks)

$$p(x) = 2x^3 - 5x^2 - 4x + 3$$

$$(x+1) \therefore a = -1 \quad \checkmark$$

$$\begin{aligned} p(-1) &= 2(-1)^3 - 5(-1)^2 - 4(-1) + 3 \quad \checkmark \\ &= -2 - 5 + 4 + 3 \quad \checkmark \\ &= 0 \end{aligned}$$

Therefore $(x+1)$ is a factor $\checkmark$

$$(x-3) \therefore a = 3 \quad \checkmark$$

$$\begin{aligned} p(3) &= 2(3)^3 - 5(3)^2 - 4(3) + 3 \quad \checkmark \\ &= 2 \times 27 - 5 \times 9 - 12 + 3 \quad \checkmark \\ &= 54 - 45 - 12 + 3 \\ &= 0 \end{aligned}$$

Therefore $(x-3)$ is a factor $\checkmark$

10. Fully factorise $x^3 - 3x^2 - x + 3$ (7 marks)

$$\text{Try } (x+1) \therefore a = -1$$

$$p(x) = x^3 - 3x^2 - x + 3$$

$$\begin{aligned} p(-1) &= (-1)^3 - 3(-1)^2 - (-1) + 3 \\ &= -1 - 3 + 1 + 3 \\ &= 0 \therefore \text{factor} \end{aligned}$$

$\checkmark \checkmark$ finding a factor

$$\begin{array}{r} x^2 - 4x + 3 \\ x+1 \overline{) x^3 - 3x^2 - x + 3} \\ \underline{x^3 + x^2} \\ -4x^2 - x \\ \underline{-4x^2 - 4x} \\ 3x + 3 \\ \underline{3x + 3} \\ 0 \end{array}$$

$\checkmark \checkmark$ Division

$$(x^2 - 4x + 3)(x+1) \quad \checkmark \text{ Stating answer}$$

$$\therefore (x-1)(x-3)(x+1) \quad \checkmark \checkmark \text{ Completing factorisation}$$

End of Validation